

Employing Deep Learning and Discrete Wavelet Transform Approach to Classify Motor Imagery Based Brain Computer Interface System

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Abstract— In the last few years, Brain Computer Interfaces (BCI) attempted the attention of many researchers. In Motor Imagery (MI)-BCI, central nervous system directly connected to a computer or an automation system. Characteristics of the electromyographic (EEG) signals are utilized in MI-BCI systems. Various techniques have been proposed to extract EEG signal characteristics during recent years. The main objective of this research is to employ an efficient deep learning approach to extract the features of EEG signals using composition of convolutional Neural Network and discrete wavelet transform utilized in the BCI system. The deep learning approach presented in this study has rarely been explored to employ for EEG features extraction. The simulation study indicates that the presented method carry out remarkable accuracy and high performance compared with conventional approaches such as support vector machine and artificial Neural Network methods and give a powerful indicative decision making aid to assist physicians in the treatment of the right and left-hand features for real time motor imagery classification system. Furthermore, the most advantages of employing the proposed method are to eliminate the feature selection level and reducing the processing cost significantly.

Keywords—Signal processing, Brain Computer Interface, Neural Network, Deep learning, Wavelet, Classify

I. INTRODUCTION

In recent years, BCIs attempted the attention of many researchers. In the BCIs after receiving and analyzing brain signals, they are converted into the commands which will be transmitted to the output devices to provide capability of performing appropriate actions, the basic components of a BCI system is illustrated in Fig. 1 [1]. Various BCI systems have been developed to be used for diverse proposes, including the assistive devices generated for the impaired people to help them to perform their daily activities; i.e. in patients suffering from cord injuries, amyotrophic lateral sclerosis, and those with less ability to control their body [2-3]. EEG is the indication of voltage variation due to ionic current produced by the neurons during activity of the brain. The advantage of using EEG is easy to operate Motor imagery. Various approaches have been found in the literature to classify and extract characteristics of EEG signals. Kernel Support Vector Machine (SVM) presented in [4] and Radial Basis Function (RBF) [5] are very common methods. Artificial Neural Network (ANN) [6-7] is very well developed method, and Random Forest (RF) [8] are too. Nonlinear Bayesian classifiers is another category which is utilized extensively for this purpose, some common methods of this category includes Bayes quadratic and Hidden Markov Model

(HMM). The next usual methods related to the nearest neighbor classifiers. In this category the common approach is k nearest neighbors (KNN) method. Employing multilayer perceptron (MLP) in neural network methods is also suggested by researchers in this field [9]. Combinations of different classification methods [10-11]. Employing effective combinational algorithms of feature extraction and classification methods provides an effective method to reduce noise during signal characteristics identification. Deep belief network (DBN) is an unsupervised learning algorithm which its core component developed based on training layer-by-layer of the network. The weights in this network for each layer are optimized and the complexity and training time completely dependent to the number of the nodes and layers of the networks [12, 13].

The target of this research is to extract features employing discrete Wavelet transform (DWT), and classified with CNN in deep learning. The most important characteristics of non-static EEG signal extraction is the ability to produce high time resolution data. The effectiveness of the presented method in this study is carried out with eighteen subjects, nine females and nine males and the trial was done in five sessions, both with feedback and without feedback.

Simulation study illustrate that the presented approach reduces the processing cost to a significant amount, and makes it possible to acquire more precise and reliable solution in the shortest time. Furthermore, the performance of feature extraction in subject to the subject is higher than session to session.

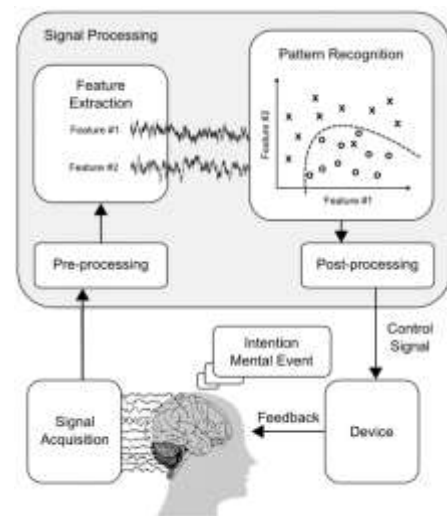


Fig. 1. Structure and configuration of a BCIs [1]

II. MATERIALS AND METHODS

A. Subjects and Data Description

The database utilized in this study is obtained from BCI competition dataset (Graz data set B). Eighteen subjects, nine females and nine males aging in the range of 21 to 35, and 21 to 27 years, respectively participated to collect data. Dataset was acquired employing multi-channel EEG and EOG signals using six electrodes with sampling rate of 250 Hz. The bipolar electrodes are installed at locations C3, Cz, C4. The standard EEG electrode placement positions in top and lateral views are depicted in Fig. 2. The trial was done in five sessions, for the first two sessions data are collected without feedback, and for the last three sessions data are collected employing feedback. The signals are pre-processed before feature extraction, using low pass and high pass filters with cut-off frequency of 100 and 0.5 Hz, respectively to remove the low and high frequency noises and continuing with notch filtering at 50 Hz to remove signal noise, next to linear regression they are classified into two classes, including left and right hand sides [14].

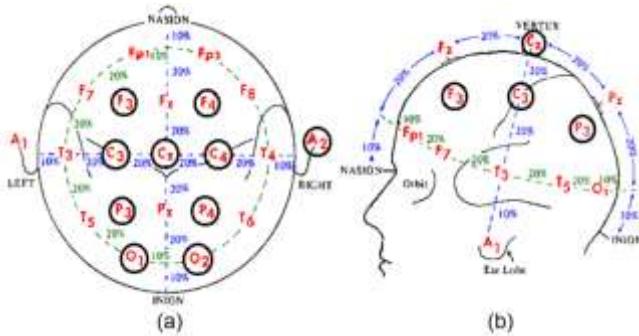


Fig. 2. The standard 10-20 electrode placement positions, (a) top view, (b) lateral view

B. Pre-Processing

Several spectral components are identified in the EEG signals. For human surface EEG signal the amplitude and its frequency ranges between 10 to 100 μ V and 0.1 to 30 Hz, respectively. The EEG signal includes various bands ranging from 0.1 to 30 Hz. The standard bands for delta signal vary between 0.1 to 3.5 Hz, theta signal include frequency range 4 to 7.5 Hz, while for the alpha signal its frequency varies between 8 to 13 Hz, and beta signal includes frequency between 14 to 30 Hz. Furthermore, gamma signals have frequency range grater than 30 Hz.

It should be noted that the noise and artifacts to be arise due to eye blink and EMG component which should be eliminated. There are two most recommended methods used for this purpose presented in the literature. Independent Component Analysis (ICA) method is used to separate a multivariable signal into its subcomponents [15]. In addition, Nonnegative Matrix Approximation (NMA) methods is utilized to factorize a matrix into the two matrices including positive elements [16]. The most recommended methods used for this purpose presented in the literature, are the Independent Component Analysis (ICA) [15] and Nonnegative Matrix Factorization (NMF) methods [16]. The comparative filter method is recommended too in the literature to determine the conversion function between each Electrooculography (EOG) and EEG recording channel. Based on previous studies, it has been shown that when the EOG components are removed

from the EEG signal, the EOG has two vertical (VEOG) and horizontal components (HEOG) [17].

In this study, data is preprocessed with high pass filtering and linear regression [18]. In addition, the Power Spectrum Density (PSD) can be employed to extract the truly EMG components.

C. Feature Extraction with Discrete Wavelet Transform

The configuration of classification procedure for EEG signals is depicted in Fig. 3. The complete procedure is defined by two stages. In the first stage, EEG signal utilized as the input to the feature extraction stage (FES), and it is transformed from time domain into the wavelet form. The output of this stage is employed as the input to the next classification stage (CS). In the first stage, the dimension of the original signal is reduced since the basic characteristics of the main signal are maintained. Accordingly, after the first stage, neural network identify the corresponding coefficients for each decomposition level.

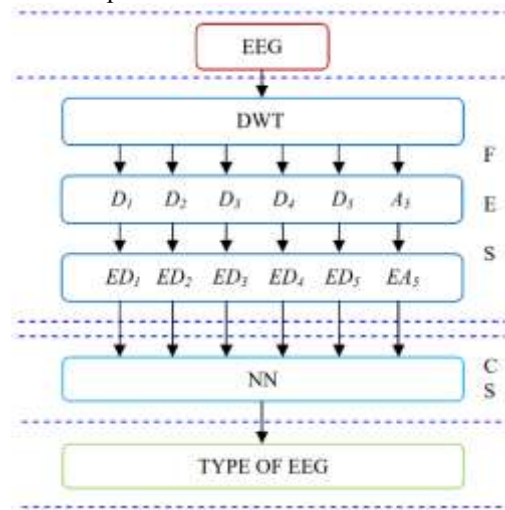


Fig. 3. Configuration of the classifier for EEG signals

The DWT has broad application in the analysis of stationary and nonstationary signals. In this transformation the signal is decomposed into finite number of time series sets, which its coefficients can be identified according to the time history of the signal in the corresponding frequency band. Two factors have vital roles in DWT, including mother wavelet function, and scaling function. Mother wavelets constructed based on one shifted function, which satisfies the high pass filtering. Following the procedure of identifying wavelet coefficients the original signal can be regenerated by the summation of the weighted wavelet functions. It should be noted that the main characteristic of DWT is its capability to localize the signal both in time and frequency domain. The mirror function improves the low pass filter, includes quadratic mirror function, which satisfied the scale function. The other advantage of using DWT signals captures the transient features. The mirror function presented by the following formula:

$$G(z)G(z^{-1})+G(-z)G(-z^{-1})=1 \quad (1)$$

$G(z)$ presents the z -transform of g .

The DWT decomposition is presented by the following equations

$$a(i)(l) = x(k) * \phi_{i,j}(k) \quad (2)$$

$$d(i)(l) = x(k) * \psi_{i,j}(k)$$

where (i) , (j) , $\phi_{i,j}(k)$ and $\psi_{i,j}(k)$ are approximation and decomposition coefficients, normalized and scale functions, respectively.

There are different wavelets tests are available. The literature review reveals that orthogonal wavelets indicate very good performance. The order of the wavelets is different and varies in this category. It is considered one, five, seven, nine, and fifteen times for Haar, Coiflets, Symmelets, Daubechies, and Biorthogonal wavelets respectively. However, Daubechies wavelet of order 4 (db4) provide an accurate choice to analyze EEG signals. Recall that the time resolution of the signal has direct relation with its frequency changes. It results decreasing the space of the wavelet coefficients. Since time and frequency features should be extracted simultaneously, it is essential to clearly identify the left and right-hand features collected in different channels.

It should be noted that for a healthy subject the total energy activity of the EEG signal in the D3 and D4 channels are similar and around 20%. For the frequency component of D5 and D2 the energy activity is limited around 10%, and 5%, respectively. Obviously, in comparison between patient and healthy subjects the energy distribution of EEG signal is different and significantly involved lower percentage. In this case, 4 channels, D3, D4, D5, and A5 were selected for decomposition. The Energy level for the last channel is weak. D1-D3 are noises, and they must be eliminated from the wavelet signal. In this case, the features can be extracted, and using Convolutional Neural Networks (CNN) classifier makes it possible to identify the right and left-hand features.

D. Convolutional Neural Network

A CNN is very efficient to perform analysis images [19]. It is broadly utilized for video analysis, visual classification, medical image and language processing [20] and specially for BCIs [21]. In this study, CNN as a deep learning method is utilized to classify the two-type of classes, including right and left-hand features. Almost too many classifiers are used for classification purpose, but the idea of using the deep learning, in BCI was not reported.

A CNN consists hidden layers which be located between an input and output layers. Convolution in the hidden layers were performed by three layers, the convolutional, pooling, and fully connected layers. In the convolutional layer, convolutional kernel in the form of small squares find out patterns by sliding over the given image. In this case, the kernel returns a number as an indication of matching or mismatching the kernel's pattern with the scanned image. To clarify, when the kernel returns a large positive value, it means that there is fully matching and when the returned value is very small or zero it means that there is no match. A deep learning classifier structure is shown in Fig. 4.

The proposed CNN in this paper contains three convolutional layers, two pooling layers, and one fully connected layer. Layers 2, 4, 6 and 8 are convolutional layers, which show the features of the right hand and left hand in the input. The output of CNN is presented by following equation:

$$C_m = \sum_{n=0}^{N-1} f_n k_{m-n} \quad (3)$$

where, C , f , k , and N are output, filter, signal, and the number of data points in k , respectively. The subscript n

illustrates the filter, and m shows the output. A pooling is employed to reduce the dimension of the features; the pooling layers are constant through the translation. Through the different kinds of pooling, max pooling is the most useful one, the maximum value with stride 2; retraining of feature map is carried out when the network perform maximum pooling. Each layer is connected to neurons in the network inside the fully connected layer. The relationship representing the output of existing layer can be given by:

$$x_i = \sum_j w_{ij} y_j + b_i \quad (4)$$

where, w , b and y illustrate the weights, bias, and the output from the previous layer, respectively, and x is the output of the existing layer.

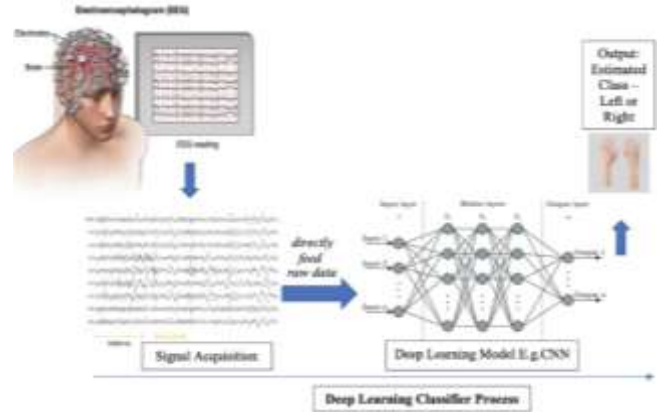


Fig. 4. The structure of EEG classifier

III. SIMULATION RESULTS

In this study, three channels $C3$, Cz , and $C4$ used for collecting of EEG signals. $C3$ channel placed at left hemisphere of the head detecting the electrical activity of right-hand motor imagery features, and $C4$ channel placed at right hemisphere of the brain detecting the electrical activity of left-hand motor imagery features. The proposed method was implemented on MATLAB software to extract motor imagery features. TABLE I. demonstrates the precision of the left-hand and right-hand features of four patients. Different classification approaches are employed and the results are compared with each other in this study. Simulation results indicate that using one channel individually may not provide results with the desired accuracy. Highest accuracy obtained using right $C3$, and Cz channels. In the proposed method, 80% of the data employed to train and 20% for test data. To train the network a direct loss minimization method is utilized to optimize the loss function. In this case, the gradient descent method is used to minimize the loss function, and the weights in each of the iterations are updated.

TABLE I. ACCURACY CHECKING OF DIFFERENT CHANNELS

	C3	C3&C4	C3&Cz	C4&Cz	Cz	C4
B01T	58.00%	66.67%	91.67%	58.4%	33.36%	41.67%
B02T	50.00%	75.00%	100%	75.00%	50.00%	50.00%
B03T	50.33%	66.67%	100%	66.66%	33.33%	41.66%
B04T	41.67%	75.00%	100%	83.33%	41.67%	83.33%

TABLE II. illustrates the compression between various methods and the proposed method, data in TABLE II indicates the output of BCI competition and simulation study presented in this paper as a function of kappa value. The advantage of

using CNN for classification is eliminating the feature selection level, however, the processing cost reduced to significant amounts. The result of the presented approach is based on feature extraction with DWT and CNN classifier is compared with the results of Chin et.al which proposed FBSP+Naïve Bayes classifier [22], Gan et.al proposed Common spatial-subspace + LDA [23] and Coyle et al Log proposed variance + LDA & SVM [24], Bentelsman suggested the RF+FBSP [25], DFFS-structure, random forest method in the lack of DFFS which the frequency of extracted features limited in 8 Hz to 30 Hz [26], and the suggested Bispectrum method [27]. Simulation results indicate that employing the CNN, the accuracy increases to the maximum value. In some subjects using the right C3 and Cz channel, the accuracy approaches to 100%. Furthermore, it should be noted that using the deep learning algorithm (CNN), the accuracy increases dramatically. The results for the subject 4, 5, 8 are close to the best, whereas, in the proposed method for the subject 2 the result is the lowest.

TABLE II. ACCURACY PERFORMANCE COMPARISON

	1	2	3	4	5	6	7	8
[22]	0.40	0.21	0.22	0.95	0.86	0.61	0.56	0.85
[23]	0.42	0.21	0.14	0.94	0.71	0.62	0.61	0.84
[24]	0.19	0.12	0.12	0.77	0.57	0.49	0.38	0.85
[25]	0.46	0.33	0.30	0.94	0.76	0.58	0.51	0.81
[26]	0.46	0.35	0.26	0.95	0.91	0.73	0.69	0.92
[27]	0.54	0.29	0.22	0.93	0.64	0.69	0.50	0.82
P.M.	0.34	0.18	0.34	0.94	0.92	0.38	0.54	0.92

P.M. = PROPOSED METHOD

IV. CONCLUSIONS

In this paper an efficient method is presented based on combination of neural network and wavelet transform to classify of right- and left-hand feature extraction. In the proposed approach four levels for decomposition of EEG signals are selected. In this case, D3 to D5 are used for detail function, and A5 for approximation. The accuracy of different channels is compared with each other, and the result shows that individual channels have the lowest accuracy in comparison with two different channels. However, the result shows that channel C3 and Cz have the maximum accuracy. To obtain the desired accuracy the combination of the channels is recommended. CNN is the fast and powerful technique for deep learning classification. One of the most advantages of employing CNN is to eliminate the feature selection level and reducing the processing cost significantly. Furthermore, employing CNN as a classifier makes it possible to utilize a powerful wavelet neural network as an efficient and reliable indication for decision making. This method is useful to assist physicians to treat the patients in real time according to the signal characteristics of right- and left-hand features.

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